

NATIONAL UNIVERSITY OF SINGAPORE

MA1513 – EXAM (90 minutes)

AY 2025/2026 Semester I

INSTRUCTIONS TO STUDENTS

- (a) There are 15 MCQ and 4 written questions in this exam with a total of 90 marks.
- (b) The MCQ should be answered and submitted through Exemplify.
- (c) For the written questions (Q16 to Q19), use the exam answer booklet to write the answers.
 - Read the instructions on the cover page of the answer booklet carefully.
 - Indicate your student number clearly on the cover page of the answer booklet.
 - Your answer booklet will be collected at the end of the exam.
 - You do not need to scan and submit the PDF version to Canvas.
- (d) Students are not allowed to exit Exemplify before the end of the exam. At the end of the 90 minutes, Exemplify will close automatically, and you should immediately stop writing.
- (e) This is an open book examination.
- (f) You may use MATLAB and any calculator. However, you should explain clearly how the answers of the written questions are obtained.

MA1513 Exam (Exemplify)

AY2025/2026 Semester I

MCQ/FIB (15 Questions; 3 marks each)

Q1. Given the general solution of a non-homogeneous system $Ax = b$:

$$\begin{cases} x = s - t + 1 \\ y = s \\ z = t \end{cases}.$$

Which of the following are true?

- A. $\begin{cases} x = t - s \\ y = t \\ z = s \end{cases}$ is a general solution of $Ax = 0$
- B. $\begin{cases} x = s - t + 1 \\ y = s + 1 \\ z = t + 1 \end{cases}$ is a general solution of $Ax = b$
- C. $\begin{cases} x = s - t \\ y = s + 1 \\ z = t + 1 \end{cases}$ is a general solution of $Ax = 0$
- D. $\begin{cases} x = 1 \\ y = t \\ z = t \end{cases}$ is a general solution of $Ax = b$
- E. $\begin{cases} x = 1 - s + t \\ y = s \\ z = t \end{cases}$ is a general solution of $Ax = b$

Q2. Given the matrix

$$A = \begin{pmatrix} 1 & 1 & * \\ a & b & * \\ b & a & * \end{pmatrix}$$

where a, b are non-zero numbers and $a \neq \pm b$.

Which of the following can be the third column of A if $\det(A) = 0$?

- A. $\begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$
- B. $\begin{pmatrix} 1 \\ a + b \\ a + b \end{pmatrix}$
- C. $\begin{pmatrix} 1 \\ a \\ b \end{pmatrix}$
- D. $\begin{pmatrix} 0 \\ b \\ a \end{pmatrix}$

E. $\begin{pmatrix} 1 \\ 2a - b \\ 2b - a \end{pmatrix}$

Q3. A is an $n \times n$ matrix such that the homogeneous system $Ax = 0$ has a non-trivial solution. Which of the following statements are true?

- A. Nullity of A is not 0.
- B. The last column of the reduced row echelon form of A must be a non-pivot column.
- C. The n rows of A are linearly dependent
- D. The eigenspace of A corresponding to eigenvalue 0 has dimension 1.
- E. There is an $n \times 1$ matrix b such that the non-homogeneous system $Ax = b$ is inconsistent.

Q4. Given the matrix

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & -1 \\ 1 & 1 & 1 \end{pmatrix}$$

Which of the following sets form a basis for the column space of A?

- A. $\left\{ \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$
- B. $\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 1 \end{pmatrix} \right\}$
- C. $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$
- D. $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \right\}$
- E. $\left\{ \begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ -1 \\ -1 \end{pmatrix} \right\}$

Q5. Which of the following sets are linearly independent?

- A. $\{(10, 11, 12), (20, 21, 22), (30, 31, 32)\}$
- B. $\{(1, 3, 2, 4), (4, 2, 3, 1), (1, 1, 1, 1)\}$
- C. $\{(0, 1, 2, 3), (0, 2, 3, 1), (0, 3, 1, 2)\}$
- D. $\{(0, 0, 0, 1), (0, 0, 1, 5), (0, 1, 5, 1), (1, 5, 1, 3)\}$
- E. $\{(2, 3, 5), (7, 11, 13), (17, 19, 23), (29, 31, 37)\}$

Q6. Which of the following linear spans are equal to $\text{span}\{(1,0,1), (0,1,0)\}$?

- A. $\text{span}\{(1,1,1)\}$
- B. $\text{span}\{(1,1,0), (0,0,1)\}$
- C. $\text{span}\{(1,0,1), (0,1,0), (1,1,1)\}$
- D. $\text{span}\{(1,0,0), (0,1,0), (0,0,1)\}$
- E. $\text{span}\{(2,1,2), (1,2,1)\}$

Q7. FIB - Let $v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 1 \end{pmatrix}$, $v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$, $v_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}$, $v_4 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ and $w = \begin{pmatrix} 3 \\ 4 \\ 5 \\ 6 \end{pmatrix}$,

The coordinate of w with respect to $S = \{v_1, v_2, v_3, v_4\}$ is given by $[w]_S = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}$ where a, b, c, d are ...

Q8. FIB - Given $u_1 = (1 \ 1 \ 0 \ 2)$, $u_2 = (2 \ 0 \ 1 \ 1)$, $u_3 = (1 \ 0 \ 2 \ 1)$.

Suppose $v = (a, b, c, -3)$ is orthogonal to u_1, u_2 and u_3 . Then a, b, c are ...

Q9. FIB - Given $T: \mathbf{R}^4 \rightarrow \mathbf{R}^3$ a linear transformation such that

$T\left(\begin{pmatrix} 3 \\ 5 \\ 2 \\ 2 \end{pmatrix}\right) = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$ and $T\left(\begin{pmatrix} 4 \\ 4 \\ 2 \\ 3 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Then $T\left(\begin{pmatrix} 1 \\ 7 \\ 2 \\ 0 \end{pmatrix}\right) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where a, b, c are ...

Q10. Suppose A is a 4×4 matrix with exactly two distinct eigenvalues and three linearly independent eigenvectors. Which of the following are true?

- A. At least one of the eigenvalues has multiplicity equal to 2
- B. One of the eigenspaces has dimension 3.
- C. One of the eigenspaces has dimension 1.
- D. We can find a 4×3 matrix P that diagonalizes A .
- E. At most one of the eigenvalues has multiplicity equal to 1.

Q11. Suppose A is a 3 x 3 diagonalizable matrix with two eigenspaces

$$E_1 = \left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \right\} \text{ and } E_2 = \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Which of the following matrices diagonalize A?

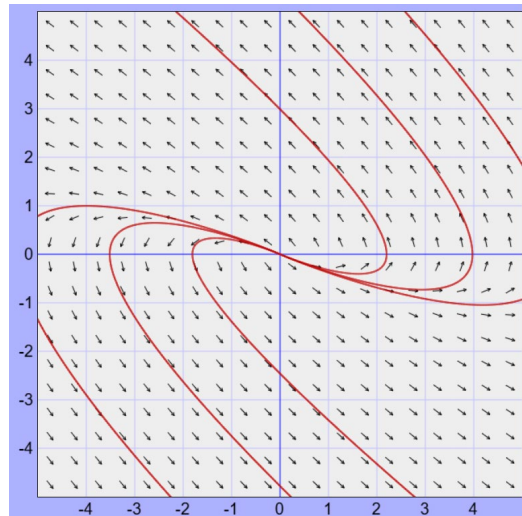
- A. $\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 4 \\ 2 & 1 & 0 \end{pmatrix}$
- B. $\begin{pmatrix} 2 & 2 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$
- C. $\begin{pmatrix} 1 & 1/2 & 0 \\ 1 & 0 & 2 \\ 1 & 1 & 2 \end{pmatrix}$
- D. $\begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 1 \\ 0 & 2 & -1 \end{pmatrix}$
- E. $\begin{pmatrix} 1 & 1 & 0 \\ 3 & 0 & 1 \\ 1 & 2 & 1 \end{pmatrix}$

Q12. An SDE has a solution given by

$$y = e^{2t} \left[\begin{pmatrix} -2 \\ 1 \end{pmatrix} \sin t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos t \right].$$

- A. The trajectories are spiralling clockwise towards the origin
- B. The trajectories are spiralling clockwise away from the origin
- C. The trajectories are spiralling anti-clockwise towards the origin
- D. The trajectories are spiralling anti-clockwise away from the origin
- E. The trajectories are close loops moving clockwise around the origin

Q13. The following is a phase portrait of a 2×2 SDE $y' = Ay$. Which of the following statements are true?



- A. The SDE has only one linearly independent solution.
- B. A has a repeated eigenvalue
- C. A has complex eigenvalues
- D. The two ends of every trajectory are parallel to the same vector
- E. There is no solution of the SDE that correspond to a straight-line trajectory

Q14. Suppose $y' = Ay$ is a 2×2 SDE and A has two eigenvalues $r_1 = 0$ and $r_2 \neq 0$ with eigenvectors $\begin{pmatrix} a \\ b \end{pmatrix}$ and $\begin{pmatrix} c \\ d \end{pmatrix}$ respectively. Which of the following are solutions of the SDE?

- A. $y = \begin{pmatrix} ka \\ kb \end{pmatrix}$ for any k
- B. $y = \begin{pmatrix} a \\ b \end{pmatrix} + t \begin{pmatrix} c \\ d \end{pmatrix}$
- C. $y = \begin{pmatrix} ce^{r_2 t} \\ de^{r_2 t} \end{pmatrix}$
- D. $y = e^{r_2 t} \begin{pmatrix} a + tc \\ b + td \end{pmatrix}$
- E. $y = \begin{pmatrix} 2a \\ 2b \end{pmatrix} - e^{r_2 t} \begin{pmatrix} c \\ d \end{pmatrix}$.

Q15. FIB - Three tanks filled with salt water are connected by pipes, where water flows in and out among the tanks at some constant rates. Let y_1, y_2, y_3 be the amount of salt in tanks 1, 2 and 3 respectively at time t . The system is described by the 3x3 SDE:

$$\begin{pmatrix} y_1' \\ y_2' \\ y_3' \end{pmatrix} = \frac{1}{100} \begin{pmatrix} -6 & 4 & 2 \\ 3 & -8 & 5 \\ 3 & 4 & -7 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}.$$

Suppose the respective amount of salt in tank 1, 2 and 3 are 10kg, 20kg and 30kg initially, and a, b, c (in kg) in the long run. Then a, b, c are ... (Type the number only, no "kg")

NATIONAL UNIVERSITY OF SINGAPORE

MA1513 Linear Algebra with Differential Equations

FINAL EXAM

Semester 1 AY2025/26

Time Allowed: 1.5 Hours

INSTRUCTIONS TO STUDENTS

1. Write down your student number on the right. Using ink or dark pencil, **completely** shade the corresponding circles for the digit or letter in each column. Cleanly erase or cover with correction tape/fluid any mistakes. DO NOT WRITE YOUR NAME!
2. This paper contains **FOUR** written questions and comprises **SIXTEEN** printed pages.
3. Students are required to answer the MCQ questions (Q1 to Q15) directly in Exemplify, and write the answers with workings of the written questions (Q16 to Q19) in this booklet.
4. This is an OPEN BOOK exam.
5. Students are allowed to use MATLAB that are installed in their PC.
6. Students are allowed to use any calculators.

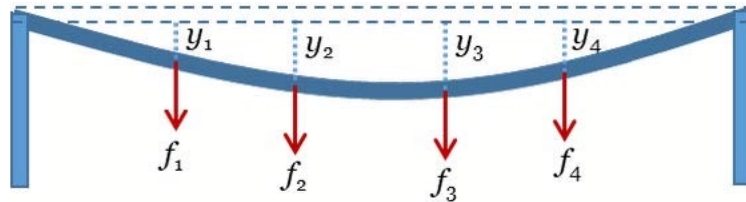
STUDENT NUMBER											
A											
U	0	0	0	0	0	0	0	0	A	N	
A	1	1	1	1	1	1	1	1	B	R	
HT	2	2	2	2	2	2	2	2	E	U	
NT	3	3	3	3	3	3	3	3	H	W	
	4	4	4	4	4	4	4	4	J	X	
	5	5	5	5	5	5	5	5	L	Y	
	6	6	6	6	6	6	6	6	M		
	7	7	7	7	7	7	7	7			
	8	8	8	8	8	8	8	8			
	9	9	9	9	9	9	9	9			

For examiner's use only		
Qns	Marks	Remarks
Q16		
Q17		
Q18		
Q19		
Total		

Instructions: Answer the questions by writing in the spaces provided in this exam answer booklet only. You may use MATLAB for all the questions. But you should show how your answers are derived.

Question 16 (10 marks)

A horizontal elastic beam is supported at each end and is subjected to downward forces f_1, f_2, f_3, f_4 (in newton) at four points as shown in the diagram, with amount of deflection y_1, y_2, y_3, y_4 (in cm) downward respectively.



The flexibility matrix is given by $D = \begin{pmatrix} 0.004 & 0.003 & 0.002 & 0.001 \\ 0.001 & 0.004 & 0.003 & 0.002 \\ 0.002 & 0.001 & 0.004 & 0.003 \\ 0.003 & 0.002 & 0.001 & 0.004 \end{pmatrix}$.

(i) (3 marks) If downward forces of 10 newton each are applied at the four points correspondingly, find the respective deflections at these points.

Answer and working

(ii) (4 marks) Is it possible to apply forces to the four points such that there are no deflection at the first three points and a deflection of 0.2cm downward at the fourth point? If yes, find the respective forces; if no, justify your answer.

Answer and working

(iii) (3 marks) Is it possible to apply two different combinations of forces f_1, f_2, f_3, f_4 at the 4 points to have the same combination of deflections y_1, y_2, y_3, y_4 at the respective points? Justify your answer.

Answer and working

Question 17 (12 marks)

Let P_1 and P_2 be two planes in the XYZ-space with equations $x + y - 2z = 0$ and $2x - y + z = 0$ respectively.

(i) (3 marks) Form a linear system with the two equations and find the solutions of the system.

Answer and working

(ii) (2 marks) Express the line of intersection (L_1) in terms of a linear span. Explain how your answer is derived.

Answer and working

(iii) (3 marks) Let P_3 be a 3rd plane with equation $x + y + z = 0$. Express P_3 in terms of a linear span. Explain how your answer is derived.

Answer and working

(iv) (4 marks) Let L_2 be the line obtained by projecting L_1 onto P_3 . Find the line (L_2) in terms of a linear span. Show your working.

Answer and working

Question 18 (10 marks)

A certain species of animals has two life stages: juvenile (up to 1 year old) and adult. Each year,

- an adult produces an average of 2 juveniles;
- half of the juveniles survive to become adults;
- 5/6 of the adults survive.

(i) (3 marks) Find the stage matrix $\mathbf{A}$ such that $\begin{pmatrix} j_{n+1} \\ a_{n+1} \end{pmatrix} = \mathbf{A} \begin{pmatrix} j_n \\ a_n \end{pmatrix}$ where j_n and a_n are the number of juveniles and adults respectively in year n .

Answer and working

(ii) (4 marks) Suppose the population of juveniles is 12000 in year 2025. What should the population of adults be in the same year so that the proportion of juveniles to adults remain the same in the following years? Explain how your answer is derived.

Answer and working

(iii) (3 marks) Will the populations of the juveniles and adults stabilize in the long run? Justify your answer.

Answer and working

Question 19 (13 marks)

The oscillation of a damped mass-string system is described by an ODE

$$my'' + cy' + ky = 0$$

where y is the (vertical) position of the ball, m is the mass of the ball in kg, k is the spring constant in Newton/meter, and c is the damping constant. Suppose the general solution of the ODE is given:

$$y = c_1 e^{-2t} + c_2 e^{-t/2}.$$

(i) (3 marks) Find the general solution of the associated SDE for this mass-spring system.

Answer and working

(ii) (2 marks) Determine the type and stability of the equilibrium point of the SDE.

Answer and working

(iii) (2 marks) Suppose the ball is displaced from its rest position, describe in words the movement of the ball before it comes to an equilibrium state.

Answer and working

(iv) (3 marks) Suppose $m = 2$. Find c and k .

Answer and working

(v) (3 marks) Sketch the phase portrait of the SDE.

Answer and working

Additional writing space

(Indicate the question and part numbers clearly.)